

Experiment 10:

7. Laboratory Exercise

A. Procedure:

a. Answer the following:

• What is a Dockerfile?

A **Dockerfile** is a text file that contains a set of instructions used to build a Docker image automatically. It defines the base image, application code, dependencies, environment variables, and commands required to run an application inside a container.

Thus, a Dockerfile acts as a **blueprint for creating Docker images** and ensures consistency across different environments.

• Explain Dockerfile commands with syntax and example.

1. FROM

- Purpose: Specifies the base image
 - Syntax:
FROM <image>:<tag>
 - Example:
FROM node:18
-

2. WORKDIR

- Purpose: Sets the working directory
 - Syntax:
WORKDIR /path
 - Example:
WORKDIR /app
-

3. COPY

- Purpose: Copies files from host to container
- Syntax:
COPY <source> <destination>

- Example:
COPY ..
-

4. RUN

- Purpose: Executes commands during build time
 - Syntax:
RUN <command>
 - Example:
RUN npm install
-

5. CMD

- Purpose: Specifies default command to run container
 - Syntax:
CMD ["executable", "arg"]
 - Example:
CMD ["node", "server.js"]
-

6. EXPOSE

- Purpose: Defines port for container
 - Syntax:
EXPOSE <port>
 - Example:
EXPOSE 3000
-

7. ENV

- Purpose: Sets environment variables
 - Syntax:
ENV key=value
 - Example:
ENV NODE_ENV=production
-

8. ENTRYPOINT

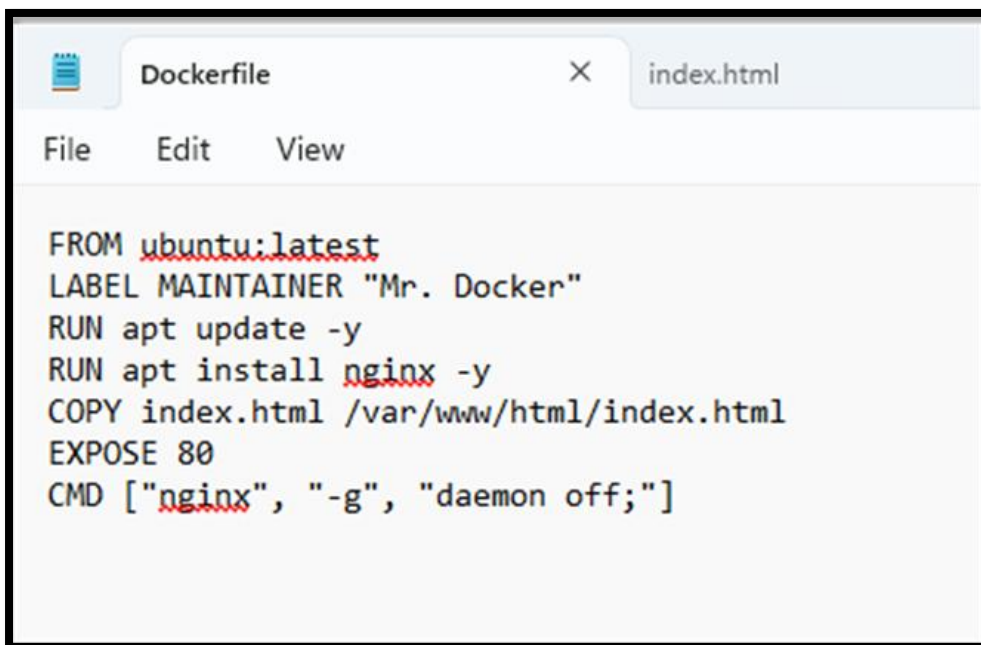
- Purpose: Sets main command (fixed execution)
 - Syntax:
ENTRYPOINT ["executable"]
 - Example:
ENTRYPOINT ["node"]
-

9. VOLUME

- Purpose: Creates persistent storage
 - Syntax:
VOLUME ["/path"]
 - Example:
VOLUME ["/app/data"]
-

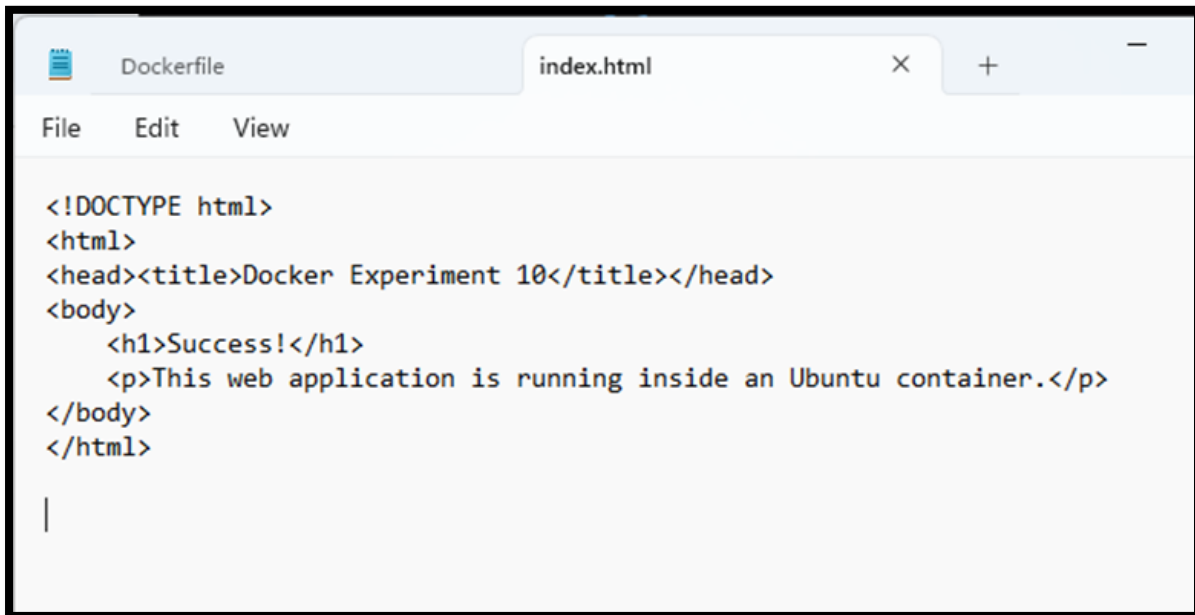
b. Execute following (Refer the shared material) and attach screenshots:

1. • Create a Dockerfile
2. • Create an html file
3. • Build and run the web application on Docker Engine

A screenshot of a code editor window. The window has two tabs: 'Dockerfile' and 'index.html'. The 'Dockerfile' tab is active, showing the following content:

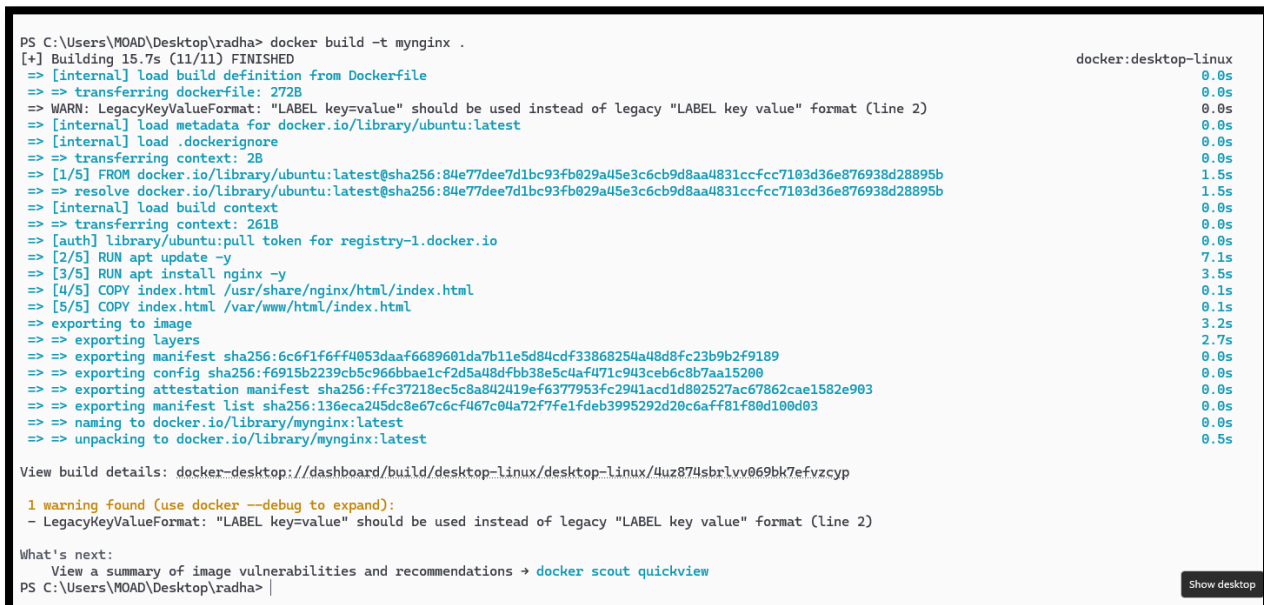
```
FROM ubuntu:latest
LABEL MAINTAINER "Mr. Docker"
RUN apt update -y
RUN apt install nginx -y
COPY index.html /var/www/html/index.html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

The 'index.html' tab is also visible but empty. The editor has a menu bar with 'File', 'Edit', and 'View' options.



The screenshot shows a web browser window with two tabs: 'Dockerfile' and 'index.html'. The 'index.html' tab is active, displaying the following HTML code:

```
<!DOCTYPE html>
<html>
<head><title>Docker Experiment 10</title></head>
<body>
  <h1>Success!</h1>
  <p>This web application is running inside an Ubuntu container.</p>
</body>
</html>
```



The screenshot shows a terminal window with the following output:

```
PS C:\Users\MOAD\Desktop\radha> docker build -t mynginx .
[+] Building 15.7s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 272B
=> WARN: LegacyKeyValueFormat: "LABEL key=value" should be used instead of legacy "LABEL key value" format (line 2)
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/ubuntu:latest@sha256:84e77dee7d1bc93fb029a45e3c6cb9d8aa4831ccfcc7103d36e876938d28895b
=> => resolve docker.io/library/ubuntu:latest@sha256:84e77dee7d1bc93fb029a45e3c6cb9d8aa4831ccfcc7103d36e876938d28895b
=> [internal] load build context
=> => transferring context: 261B
=> [auth] library/ubuntu:pull token for registry-1.docker.io
=> [2/5] RUN apt update -y
=> [3/5] RUN apt install nginx -y
=> [4/5] COPY index.html /usr/share/nginx/html/index.html
=> [5/5] COPY index.html /var/www/html/index.html
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:6c6f1f6ff4053daaf6689601da7b11e5d84cdf33868254a48d8fc23b9b2f9189
=> => exporting config sha256:f6915b2239cb5c966bbae1cf2d5a48dfbb38e5c4af471c943ceb6c8b7aa15200
=> => exporting attestation manifest sha256:ffc37218ec5c8a842419ef6377953fc2941acd1d802527ac67862cae1582e903
=> => exporting manifest list sha256:136eca245dc8e67c6cf467c04a72f7fe1fdeb3995292d20c6aff81f80d10d03
=> => naming to docker.io/library/mynginx:latest
=> => unpacking to docker.io/library/mynginx:latest

View build details: docker://dashboard/build/desktop-linux/desktop-linux/4uz874sbrlv069bk7efvzcyp

1 warning found (use docker --debug to expand):
- LegacyKeyValueFormat: "LABEL key=value" should be used instead of legacy "LABEL key value" format (line 2)

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview
PS C:\Users\MOAD\Desktop\radha>
```

```
PS C:\Users\MOAD\Desktop\shivam> docker run -d -p 6060:80 --name experiment-final mynginx
fa05f7454902f03cb6fe7dd42d71df998fa8e3e791692555a66ffb868ffcaf1d
PS C:\Users\MOAD\Desktop\shivam>
```

```
PS C:\Users\MOAD\Desktop\radha> docker run -d -p 6060:80 --name experiment-final mynginx
fa05f7454902f03cb6fe7dd42d71df998fa8e3e791692555a66ffb868ffcaf1d
PS C:\Users\MOAD\Desktop\radha>
```



The screenshot shows a web browser window with the address bar set to 'localhost:6060'. The page displays the following content:

Success!

This web application is running inside an Ubuntu container.

Containers

Images

Volumes

Builds

Docker Scout

Extensions

Containers

Container CPU usage

0.00% / 1200% (12 CPUs available)

Container memory usage

8.19MB / 7.46GB

Show charts

Search

Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	☐ pedantic_herschel	eba10fb82e25	crishanginx:<none>	8888:80	0%	1 day ago	
☐	☐ tender_volhard	e77ff8a5bb7a	crishanginx:<none>	8888:80	0%	1 day ago	
☐	☐ wonderful_roentgen	8b94a253ea2b	crishanginx:<none>	8888:80	0%	1 day ago	
☐	☐ modest_germain	1900d731f95f	crishanginx:<none>	8888:80	0%	1 day ago	
☐	☐ tender_shamir	d411e614ff13	crishanginx:<none>	8888:80	0%	1 day ago	
☐	☐ sharp_lamport	502f0fd17f81	crishanginx:<none>	8888:80	0%	1 day ago	
☐	☐ clever_brahmagupta	e5dd71aa6124	ubuntu:<none>	-	0%	2 days ago	
☐	☐ Paarth	0bd52ab082b3	ubuntu:<none>	-	0%	7 days ago	
☐	☐ charming_nash	b9173483316e	ubuntu:<none>	-	0%	5 months ago	
☐	☐ pensive_wescoff	dec01bc61541	ubuntu:<none>	-	0%	5 months ago	
☐	☐ quizzical_goldstine	ec7c1273c9f5	ubuntu:<none>	-	0%	1 hour ago	
☐	☒ experiment-final	fa05f7454902	mynginx:<none>	6060:80	0%	3 minutes ago	

Showing 18 items

Engine running

RAM 1.18 GB CPU 0.08% Disk --- GB avail. of --- GB

BETA Terminal New version available